

PhysioTwin: Teaching, Verifying, and Grading Robotic Rehabilitation Exercises with a Single Consumer Smartwatch

Firaol Tesfaye
Higher Colleges of Technology
Dubai, UAE
h00539876@hct.ac.ae

Milki Teshome
Higher Colleges of Technology
Dubai, UAE
h00539899@hct.ac.ae

Moez Rehman
Higher Colleges of Technology
Dubai, UAE
mrehman1@hct.ac.ae

Yassine Benachour
Higher Colleges of Technology
Dubai, UAE
ybenachour@hct.ac.ae

Abstract—Robot-assisted therapy can guide an impaired upper limb through prescribed exercises, but programming patient-specific trajectories today requires motion-capture laboratories or hands-on kinesthetic teaching. We present PhysioTwin, a pipeline in which a single consumer smartwatch is the entire interface to a rehabilitation robot: a demonstration recorded on the wrist teaches the robot an exercise, and the same sensing pipeline later verifies and grades performance. The system builds on a new dataset of 243 sessions from 11 healthy subjects performing 12 prescribed upper-limb exercises, most recorded simultaneously on both wrists. With it we provide the first ground-truth validation of the bilateral mirror-symmetry assumption; expose a severe cross-wrist deployment gap in exercise recognition and close it with a data-certified mirror transformation; learn exercises as dynamic movement primitives and execute them on a physics-simulated UR5e; close the loop by recognizing the simulated robot’s executions through a virtual watch; and grade movement quality against population and personalized motion envelopes. Subject-independent recognition exceeds 92%, and mirror-aware training restores cross-wrist accuracy to the within-wrist level. All experiments are reproducible from the accompanying repository.

Index Terms—rehabilitation robotics, wearable sensors, smartwatch, learning from demonstration, dynamic movement primitives, human activity recognition

I. Introduction

Musculoskeletal disorders of the shoulder, elbow, and wrist are among the most common reasons for physiotherapy referral, and their treatment depends on patients performing prescribed exercises correctly and consistently at home. Adherence is notoriously poor and largely invisible to the clinician: between appointments, therapists typically rely on patient self-reports. Two technology families address parts of this problem. Wearable-sensor systems can recognize and count exercises from a wrist-worn inertial measurement unit (IMU) [1], giving clinicians objective adherence data — a rapidly growing field surveyed in [12]. Rehabilitation robots can physically guide an impaired limb through its prescribed motion with assist-

as-needed control [5], [16], taking over the repetitive guidance work of the therapist. The two families, however, have remained separate: the robots are programmed through motion-capture laboratories, teach-pendant programming, or hands-on kinesthetic teaching — all of which require the therapist and the robot to share a room and specialized equipment.

This paper connects the two families with the cheapest sensor a patient already owns. In the proposed pipeline, which we call PhysioTwin, a demonstration of a prescribed exercise — recorded by an ordinary smartwatch on the demonstrator’s wrist, anywhere — is converted into a robot-executable motion primitive. A compliant robot arm can then guide the patient’s limb through that exercise; the clinical motivation follows mirror therapy [7], [8], in which the movement of the unaffected limb serves as the template for the affected one. The same smartwatch pipeline subsequently verifies what was executed and grades its quality. One consumer device is thus the robot’s teacher, the exercise’s examiner, and the patient’s monitor.

The work is grounded in a new dual-wrist dataset (Section III) whose distinguishing property is simultaneity: subjects performed bilateral exercises wearing a watch on each wrist, so every left-wrist recording has a synchronized right-wrist counterpart. This enables experiments that were previously impossible, most importantly a ground-truth test of the mirror-symmetry assumption that underlies both mirror therapy instrumentation and left/right data augmentation in wearable machine learning.

The contributions are:

- 1) the first ground-truth validation of bilateral mirror symmetry from synchronized dual-wrist consumer smartwatches, quantified per exercise (Section V-A);
- 2) the first measurement of the cross-wrist deployment gap for clinical exercise recognition (a 35–40 percentage-point collapse) and its complete closure with a data-certified mirror transformation, evalu-

ated under the strictest unseen-subject-and-unseen-wrist protocol (Section V-B);

- 3) an end-to-end, reproducible pipeline from smartwatch demonstration to robot execution — repetition segmentation, dynamic movement primitive (DMP) learning, kinematic retargeting to a UR5e, and physics-simulated execution through the standard ROS 2 control stack (Sections V-C, V-D);
- 4) a closed verification loop in which a classifier trained exclusively on human recordings recognizes the robot’s executions through a virtual watch (Section V-E); and
- 5) a movement-quality grading scheme based on population and personalized motion envelopes, evaluated on 2,452 genuine repetitions with controlled fault injection (Section V-F).

All experiments are healthy-subject and simulation-based by design; Section VI discusses this scope honestly.

II. Related Work

Wearable exercise recognition. Burns et al. [1] established smartwatch-based recognition of shoulder physiotherapy exercises; subsequent work has raised accuracy with deep models and larger exercise vocabularies. Feature-engineered classical pipelines remain competitive at small subject counts; the present dataset’s pilot predecessor was analyzed this way by Benachour et al. [2], who optimized windowing, feature selection, and classical models on three movements from three subjects. Follow-up studies from the same group compared classical models and an RNN for wearable rehabilitation movement classification with Core ML on-watch deployment analysis [3], and proposed a hybrid CNN–LSTM that, notably, closed its subject-independent generalization gap by fine-tuning on roughly 60 s of user calibration data [4] — independently corroborating the value of short calibration sessions that our personalized quality envelope exploits (Section V-F). Parallel work in the group explores dynamic-time-warping representations for upper-limb IMU classification — combinatorial DTW medoid distances [14] and prototype-based DTW features [15] — an alternative movement-representation route to the windowed statistical features used here. We adopt this line’s windowing/feature philosophy for our recognition baseline but address questions that require the new dataset’s scale and dual-wrist design.

Sensor placement and the wearing-side problem. Cross-position generalization is a recognized open problem in human activity recognition [9]. For the specific left/right wrist question, prior work has simulated the opposite wrist by mirroring signal axes; to our knowledge no public dataset provides synchronized recordings of the same movement from both wrists, so the mirror assumption itself has not previously been testable.

Robot-assisted upper-limb therapy. From the pioneering MIT-MANUS trials [16] to today’s end-effector and ex-

oskeleton devices [5], robot-assisted upper-limb therapy is clinically established, and assist-as-needed control is standard. Exercise content, however, is programmed through preinstalled libraries, kinesthetic teaching, or motion capture. Learning from demonstration [17], notably with DMPs [6], is a mature robotics technique, and IMU-driven demonstration has been explored for reaching motions; a consumer smartwatch as the sole teaching interface for rehabilitation exercises has not been reported.

Mirror therapy. Using the healthy limb’s motion as the template for the impaired one has clinical foundations in mirror-box therapy [7] and meta-analytic support in stroke rehabilitation [8]. PhysioTwin can be read as a robotic instantiation: the healthy arm demonstrates, the robot reproduces the mirrored motion on the affected side.

Movement quality. Smoothness metrics, notably log dimensionless jerk [10], are validated proxies of motor performance. Learned anomaly detectors have also been applied to rehabilitation monitoring — e.g., a hybrid multi-task LSTM autoencoder for semantic anomaly diagnosis [13] — whereas we deliberately choose interpretable per-repetition envelopes (range, tempo, spectral, and end-point features), whose individual bounds name the fault they flag; the two approaches are complementary.

III. Dataset

A. Participants and ethics

Eleven healthy adult university students (anonymized S01–S11) volunteered for the recordings. Inclusion required being an adult with no current upper-limb injury, pain, or movement restriction; participants with any musculoskeletal complaint affecting the recorded movements were excluded, and all recordings are labelled as performed with unaffected limbs. All participants gave informed consent, and identities are anonymized in all released code and results.

B. Acquisition setup

Recordings were made with two Apple Watch units (watchOS 26.2–26.3) running a custom watchOS acquisition application; the same two physical devices were used for all subjects, one always on the left wrist with the crown facing right and the other always on the right wrist with the crown facing left (watch face dorsal, standard wear). Each session logs the CoreMotion device-motion service at a nominal 100 Hz: attitude quaternion in the XArbitraryCorrectedZVertical reference frame (gravity-aligned z , arbitrary heading), rotation rate (rad/s), user acceleration and gravity (units of g), alongside the raw accelerometer stream and a per-session metadata file. No user-side calibration is performed beyond CoreMotion’s internal sensor fusion. Every sample carries a Unix-epoch millisecond timestamp from the watch’s NTP-disciplined clock, which synchronizes the two devices; residual offset is absorbed in analysis by a ± 1 s cross-correlation lag

search. Effective sampling rates span 99.5–100.2 Hz with no missing samples.

C. Exercise protocol

Each subject performed 12 prescribed upper-limb rehabilitation exercises associated with seven musculoskeletal conditions (e.g., frozen shoulder, shoulder impingement, distal radius fracture): bicep curl, external rotation, forearm pronation–supination, front and lateral deltoid raises, scaption, scapular retraction, ulnar nerve glide, wand flexion, wand extension, banded overhead reach, and one-arm tricep extension. All subjects followed the same protocol: continuous self-paced repetitions for a fixed per-exercise recording duration (30 s for the wand exercises, 90 s for the ulnar nerve glide, 40 s otherwise), yielding a median of 10 repetitions per session with no rest periods within a session. This gives 243 sessions and roughly 2.7 hours of motion. The property central to this paper is that ten of the twelve exercises were performed bilaterally with a watch on each wrist: of 110 left/right session pairs, 108 overlap in absolute time — synchronized recordings of the same physical movement seen from both wrists (the remaining two exercises are single-arm by design).

IV. Methods

A. Frame-invariant motion representation

The two watches occupy different body frames (opposite wrists, opposite crown orientations), and each device-motion stream has an arbitrary heading reference. All cross-wrist comparisons therefore use quantities invariant to the sensor frame. From the attitude quaternion sequence $q(t)$, resampled onto a common 100 Hz grid by spherical linear interpolation, we compute the rotation angle relative to the session’s initial pose, $\theta(t) = \angle(q_{\text{ref}}^{-1}q(t))$, and the angular speed $\|\omega(t)\|$ from the gyroscope. Residual clock offset between watches is absorbed by a ± 1 s lag search.

B. Mirror validation

For each synchronized left/right pair we correlate the two wrists’ $\theta(t)$ profiles (Pearson r , after lag alignment) and report the RMSE in degrees. Because $\theta(t)$ is frame-invariant, agreement directly measures bilateral motion symmetry rather than sensor alignment.

C. Exercise recognition and the mirror map

Recognition follows the standard windowing pipeline: 2.56 s windows with 50% overlap; per window, 12 time- and frequency-domain statistics for each of 13 channels (user acceleration, rotation rate, gravity, pitch, roll, and the two frame-invariant magnitudes), for 156 features; a 300-tree random forest; and strict leave-one-subject-out (LOSO) evaluation. Absolute yaw is excluded because its arbitrary per-session reference would leak session identity. Session-level decisions use majority voting.

For cross-wrist transfer we estimate a data-certified mirror map: for every channel, the sign relating left- and

right-wrist signals is taken as the sign of the median correlation across the synchronized pairs. Training data are then mirrored through this map (alone or as augmentation), and evaluation includes the strictest protocol in which the test subject and the test wrist are both unseen. To rule out leakage through the transformation itself, the map was re-estimated within every LOSO training fold (i.e., excluding the held-out subject’s pairs): the resulting sign patterns are identical to the global map in all 11 folds, so no information from the test subject enters the mirror transformation. No other preprocessing uses dataset-level statistics — features are computed per window, and random-forest training requires no global normalization.

D. Repetition segmentation and primitive learning

Repetitions are segmented from the smoothed $\theta(t)$ profile by peak detection (prominence 40% of range) with boundaries at inter-peak minima. Each session’s median-duration repetition serves as the demonstration and is encoded as a discrete DMP [6] (50 radial basis functions per dimension) on the heading-normalized rotation vector. DMP time and goal parameters provide the clinically required scaling knobs: tempo (τ) and range of motion (goal).

E. Retargeting and execution

The robot reproduces the hand path rather than imitating human anatomy: a virtual elbow is anchored a forearm’s length (0.25 m) above the UR5e’s home tool pose, and the tool tracks $p(t) = a + R(t)v_0$, $R_{\text{tool}}(t) = R(t)R_0$, where $R(t)$ is the demonstrated wrist rotation re-referenced to the repetition start. Joint trajectories are obtained by damped least-squares inverse kinematics with singularity-adaptive damping and step capping, checked against UR5e joint and speed limits, and executed in Gazebo through the ROS 2 joint_trajectory_controller as five-repetition therapy sets with 2 s return-to-start blends, at therapy pace (at least half the demonstration speed, slower where speed limits require).

F. Virtual-watch verification and quality grading

A virtual watch is strapped to the executed motion: from recorded joint states, forward kinematics reconstructs the tool rotation, which is re-expressed in the demonstrating watch’s frame (composing the session reference and repetition-start attitudes) and speed-compensated back to demonstration tempo; gravity, rotation rate, user acceleration, pitch, and roll are then synthesized and passed through the unchanged recognition pipeline.

Quality grading summarizes each repetition by six interpretable features: range of motion, active duration, range-normalized peak speed, log dimensionless jerk [10], high-frequency (>3 Hz) velocity power, and end-point (return-to-start) error. An exercise’s envelope is a robust interval

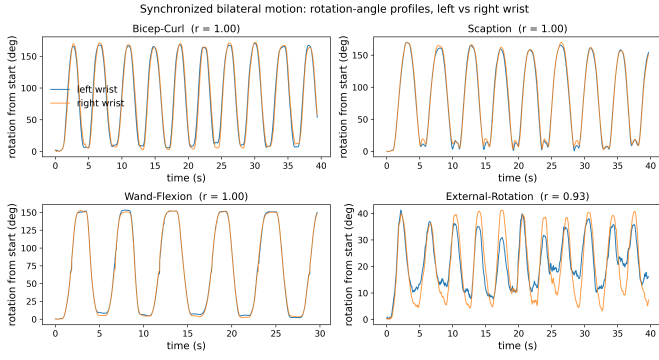


Fig. 1. Synchronized bilateral motion: frame-invariant rotation-angle profiles of the left (blue) and right (orange) wrists during four exercises of one subject.

(median $\pm K \cdot \text{MAD}$, per-feature K) learned from genuine repetitions; bounds are one-sided where only one direction is clinically abnormal. Faults are injected into genuine human repetitions — $2\times$ speed, half range, $4 \text{ Hz}/3^\circ$ tremor, truncation at 60% — and detection is evaluated LOSO. A personalized envelope centers the intervals on a subject’s own calibration repetitions (every second repetition) with the population width as a floor. Operationally this corresponds to a brief supervised calibration — a handful of correctly performed repetitions recorded under clinician observation at the start of a treatment block — after which unsupervised home sessions are graded against that patient’s own baseline; the population floor guards against the small calibration set under-estimating natural variability.

V. Results

A. Bilateral mirror symmetry (Experiment 1)

Across 107 synchronized pairs, 9 of 10 dual-wrist exercises show left/right rotation-profile correlations of $r \geq 0.97$ (most 0.99–1.00) with 4–16° RMSE (Fig. 1). External rotation is the outlier ($r = 0.75 \pm 0.16$): its rotation amplitude is small ($\approx 40^\circ$ versus up to 170° for curls) and genuine inter-arm asymmetry is visible in the profiles. Bilateral mirror symmetry — previously an assumption — is thus quantified: excellent for gross exercises, measurably imperfect for subtle ones.

B. Recognition and the cross-wrist gap (Experiments 2, 2b)

The LOSO baseline reaches 92.1% window and 93.7% session accuracy (macro-F1 0.916) over 12 classes and 8,187 windows (Fig. 2); residual confusion concentrates among the three small shoulder-elevation movements (lateral deltoid raise, scapular retraction, front deltoid raise). Transferring across wrists without correction collapses accuracy to 59.5% (train right, test left) and 52.9% (reverse). The estimated mirror map is physically coherent (gravity x/z , roll, and gyro y transfer directly; pitch,

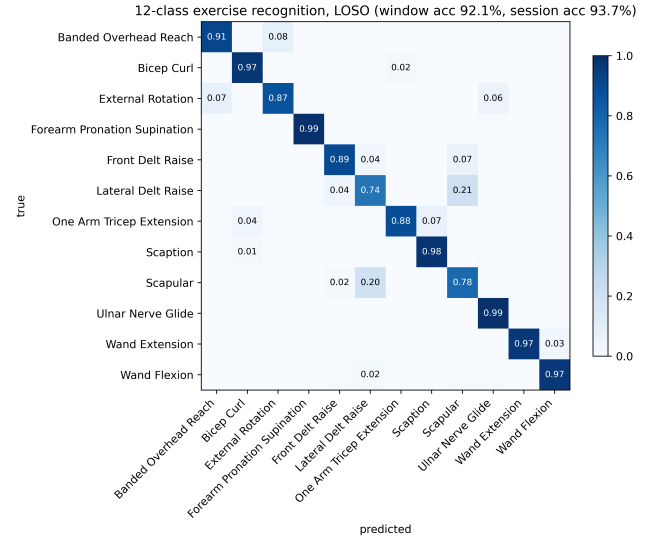


Fig. 2. 12-class LOSO confusion matrix (window level, all folds).

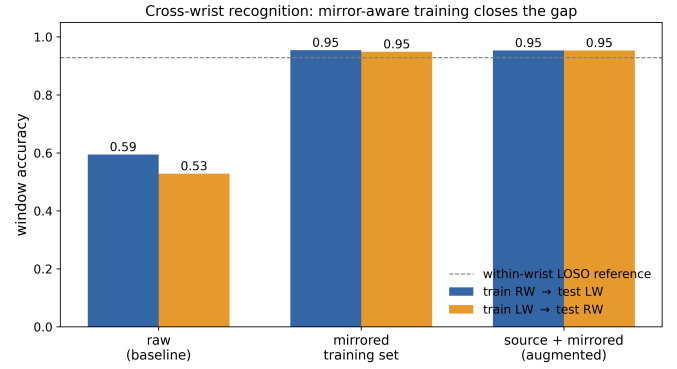


Fig. 3. Cross-wrist recognition with raw, mirrored, and augmented training. The dashed line is the within-wrist LOSO reference.

gravity y , and gyro x/z flip), and training on mirror-transformed data restores 94.9–95.5%. Under the strictest protocol — unseen subject and unseen wrist — mirror-augmented training achieves 92.5–93.2%, statistically indistinguishable from the within-wrist reference (92.9%): the deployment gap is closed (Fig. 3).

C. Repetitions and motion primitives (Experiment 3)

Segmentation yields 2,452 repetitions across 239 of 240 sessions. Because the dual-wrist recordings observe the same physical movement, per-wrist repetition counts validate each other: counts agree exactly in 86.9% of pairs and within ± 1 in 94.4%, with five of the six larger disagreements again on external rotation. Per-exercise DMPs reproduce their demonstrations with 0.5–1.7° median RMSE, and the same primitive replays at scaled tempo and range (Fig. 4) — one demonstration yields a full difficulty ladder.

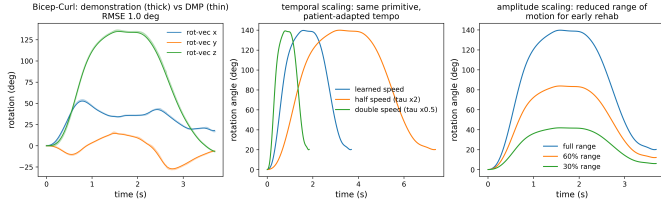


Fig. 4. Left: DMP reconstruction over the demonstration. Middle, right: temporal and amplitude scaling of one learned primitive.

D. Robot execution in simulation (Experiments 4a, 4b)

All 12 retargeted exercises track their pose references at sub-millimeter/sub-0.25° IK error within UR5e joint limits. Nine execute at demonstration speed; forearm pronation-supination, ulnar nerve glide, and wand flexion exceed wrist speed limits at human tempo and require DMP time scaling of 2.05–3.02× — the primitive representation supplies exactly this knob. In Gazebo, all 12 exercises execute as five-repetition therapy sets with 0.07–0.63° mean joint tracking error.

E. Closed-loop verification in simulation (Experiment 4c)

A random forest trained on all human windows classifies the virtual-watch signals of the robot’s executions correctly for 10 of 12 exercises, most with 100% window-vote agreement. The two failures (front deltoid raise → scapular; lateral deltoid raise → front deltoid raise) fall inside the classifier’s known human confusion cluster, amplified by the single-pivot retargeting model, which most distorts translation-dominant raises. Two negative results proved diagnostic: single-repetition executions were almost universally misclassified (the features encode repetition rhythm), and omitting the session’s absolute reference attitude from the virtual watch shifted all gravity-dependent features (2/12 until corrected).

F. Movement-quality grading (Experiments 5, 5b)

On 2,452 genuine repetitions (LOSO), the population envelope flags healthy repetitions at 8.3% while detecting injected tremor at 90%, truncation at 95% (via end-point error), and half-range at 68%. Tempo faults evade it (23%): healthy tempo varies more than 2× across people, so a population bound cannot represent “too fast for you”. The personalized envelope (own calibration repetitions as center, quarter population width as floor) lifts tempo detection to 66% and half-range to 83% at 7.7% false alarms — a short supervised calibration session buys personalized home grading (Fig. 5).

Executed on the robot (three exercises, five variants each), the grader behaves consistently with the offline study. Tremor is flagged on 15 of 15 repetitions with the correct feature named (hf_ratio); reduced range on 10 of 10 repetitions for the two exercises with narrow population ROM spread (rom_deg), while wand extension’s wide spread defeats the population bound — matching its

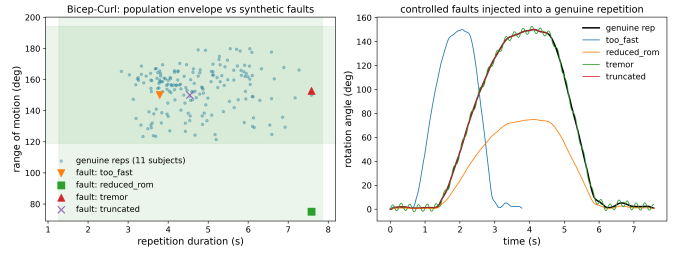


Fig. 5. Left: population envelope (shaded) for one exercise with fault positions of one repetition. Right: the injected fault types.

offline rate. Clean executions pass, apart from marginal duration flags on one exercise traced to the inter-repetition return blend merging into the repetition segment. Over-speed programs are caught before grading: at full demonstration tempo the trajectory controller aborts on path tolerance — an execution-layer safety net — while over-speed inside tracking tolerance passes the population envelope, consistent with the offline finding that tempo faults require the personalized envelope. Truncated programs are not flagged because they cannot physically occur in guided mode: the protocol’s return-to-start blend completes every repetition by design; truncation remains detectable (95%) for free-moving patients.

VI. Discussion and Limitations

Scope. We emphasize that every robot result in this paper is simulation-based: no physical robot was operated, and nothing here constitutes physical-robot validation or clinical deployment. Likewise, all subjects are healthy adults, and the movement faults of Section V-F are controlled synthetic injections rather than clinician-labelled patient errors — detection rates on real pathological movement may differ and require clinical data to establish. The paper claims a validated pipeline, not a clinical outcome. Future physical execution is intended to layer admittance-controlled guidance over the collaborative arm’s built-in safety envelope — enforced joint position and velocity limits (the trajectories here already respect UR5e limits by construction), reduced-speed mode, and a tool-center-point contact force limit on the order of tens of newtons, well below injury thresholds — so that the software trajectory is bounded by hardware-enforced limits rather than trusted alone. Any human-robot therapy additionally requires ethics approval and a formal risk assessment; real human-robot therapy remains future work.

The external-rotation thread. The same exercise is the outlier in mirror symmetry, repetition-count agreement, and (in human data) recognition confusions: small-amplitude movements stress every stage of a wrist-only pipeline. A second sensing modality or an upper arm placement may be required for such exercises.

Retargeting fidelity. The single-pivot forearm model reproduces hand orientation arcs faithfully but distorts translation-dominant movements, which is visible as the

two verification failures. A richer target would add an estimated shoulder joint center and a subject-scaled two-segment (upper-arm/forearm) chain, and could fuse the watch's linear user-acceleration channel — unused here — to constrain the hand's translational path rather than its orientation alone. This is a natural next step precisely because the failing exercises (deltoid raises) are the ones where hand translation, not wrist orientation, carries the movement.

Grading robot motion with human norms. Idealized primitives are smoother than any human; a naive envelope flags them for it. Clinically one-sided bounds (smoother-than-human is not a fault) resolved this, a caution for anyone applying human-derived norms to robot-mediated therapy.

VII. Conclusion

A single consumer smartwatch can teach a physics-simulated rehabilitation robot a 12-exercise clinical repertoire, verify its execution, and grade movement quality — with the left/right mirror assumption underpinning wearable rehabilitation finally measured rather than assumed, and the cross-wrist deployment gap it explains closed completely. The dual-wrist dataset, all analysis code, and every figure in this paper are reproducible from the public repository (https://github.com/Firaol-Tes/ASET_REHAB). Future work: execution on a physical UR5e, admittance-controlled guidance, patient populations with clinical partners, and richer arm models for translation-dominant exercises.

Data Availability

All analysis code, experiment scripts, anonymized per-session results, and the source of every figure and table in this paper are publicly available at https://github.com/Firaol-Tes/ASET_REHAB. The raw sensor recordings contain participant identifiers and are available from the authors on reasonable request, subject to the consent terms of the collection.

Acknowledgments

The authors thank the volunteer subjects who contributed the recordings.

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